6/1/26

Unit-1 : Matrices

System of Linear equation

An eq^n with variables $x_1, x_2, x_3 \ldots x_n$ is called a linear eq^n if the variables have a degree 1. The collection of all linear eq^{n}s is called system of linear eq^n. The General form is,

$$a_{11}x_1 + a_{12}x_2 + a_{13}x_3 + \ldots + a_{1n}x_n = b_1$$
$$a_{21}x_1 + a_{22}x_2 + a_{23}x_3 + \ldots + a_{2n}x_n = b_2$$
$$\vdots$$
$$a_{m1}x_1 + a_{m2}x_2 + a_{m3}x_3 + \ldots + a_{mn}x_n = b_m$$

The Matrix form is

$$\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & & a_{mn} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix}$$

$$AX = B$$

where A is coefficient matrix, X is unknown matrix and B is constant matrix.

NOTE: ① If the system of eq^n is of the form $AX = O$ (Matrix $B = O$) is called Homogenious SLE. Otherwise (Matrix $B \neq O$) is called Non-HSLE

② The 1st non-zero element of the row is called the leading entry or pivot. (LE)

③ The positions of pivot elements is called pivot position.

④ The columns in which pivot elements are present are pivot columns.

Elementary Transformation Matrix

1] ROW

a) Interchanging 2 rows $(R_i \leftrightarrow R_j)$
b) Multiplying all the entries in a row with non-zero constants $(R_j \rightarrow kR_j)$
c) Replace 1 row by sum of itself and multiple of another row and a non-zero constant $(R_i \rightarrow R_i + kR_j)$

Echelon form of Matrix

We can say that given matrix is in echelon form if it satisfies following properties:

1] The L.E of any row must be rightside of LE of previous row.

2] The elements below the LE (in the column) must be 0.

3] All the zero rows must be (the) at the bottom of the matrix

26 ## Rank of a Matrix

No. of Non zero rows in the echelon form of the Matrix denoted by $\rho(A)$.

Ex: $A = \begin{bmatrix} 2 & 3 & 4 \\ 0 & 5 & 6 \\ 0 & 0 & 0 \end{bmatrix}$, $\rho(A) = 2$

Sums

1. Find Rank.

a) $A=\begin{bmatrix} 1 & 2 & -1 & 3 \\ 3 & 4 & 0 & -1 \\ -1 & 0 & -2 & 7 \end{bmatrix}$

$\Rightarrow$ I. $R_3 \to R_3 + R_1$
$R_2 \to R_2 - 3R_1$
II. $R_3 \to 8R_3 + R_2$

I. $A \sim \begin{bmatrix} 1 & 2 & -1 & 3 \\ 0 & -2 & 3 & -10 \\ 0 & 2 & -3 & 10 \end{bmatrix}$

II. $A \sim \begin{bmatrix} 1 & 2 & -1 & 3 \\ 0 & -2 & 3 & -10 \\ 0 & 0 & 0 & 0 \end{bmatrix}$ $\therefore \underline{\rho(A) = 2}$

b) $A=\begin{bmatrix} 3 & -3 & 0 \\ 1 & 4 & 5 \\ 4 & 4 & 8 \end{bmatrix}$

$\Rightarrow$ I. $R_2 \to 3R_2 - R_1$ II. $R_3 \to 5R_3 - 8R_2$
$R_3 \to 3R_3 - 4R_1$

$A \sim \begin{bmatrix} 3 & -3 & 0 \\ 0 & 15 & 15 \\ 0 & 24 & 24 \end{bmatrix}$ $A \sim \begin{bmatrix} 3 & -3 & 0 \\ 0 & 15 & 15 \\ 0 & 0 & 0 \end{bmatrix}$

$\therefore \underline{\rho(A) = 2}$

c) $A = \begin{bmatrix} 1 & 2 & 3 & 0 \\ 2 & 1 & 3 & 2 \\ 3 & 2 & 1 & 3 \\ 6 & 8 & 7 & 5 \end{bmatrix}$

I. $R_2 \to R_2 - 2R_1$
$R_3 \to R_3 - 3R_1$
$R_4 \to R_4 - 6R_1$

$A \sim \begin{bmatrix} 1 & 2 & 3 & 0 \\ 0 & 0 & -3 & 2 \\ 0 & -4 & -8 & 3 \\ 0 & -4 & -11 & 5 \end{bmatrix}$

II. $R_2 \leftrightarrow R_3$

$A \sim \begin{bmatrix} 1 & 2 & 3 & 0 \\ 0 & -4 & -8 & 3 \\ 0 & 0 & -3 & 2 \\ 0 & -4 & -11 & 5 \end{bmatrix}$

III. $R_4 \to R_4 - R_2$

$A \sim \begin{bmatrix} 1 & 2 & 3 & 0 \\ 0 & -4 & -8 & 3 \\ 0 & 0 & -3 & 2 \\ 0 & 0 & -3 & 2 \end{bmatrix}$

IV. $R_4 \to R_4 - R_3$

$A \sim \begin{bmatrix} 1 & 2 & 3 & 0 \\ 0 & -4 & -8 & 3 \\ 0 & 0 & -3 & 2 \\ 0 & 0 & 0 & 0 \end{bmatrix}$

$\rho(A) = 3$

d) $A = \begin{bmatrix} 2 & -1 & 3 & -1 \\ 1 & 3 & 3 & -1 \\ 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & -1 \end{bmatrix}$

I. $R_2 \longrightarrow 2R_2 - R_1$
$R_3 \longrightarrow 2R_3 - R_1$

$A \sim \begin{bmatrix} 2 & -1 & -3 & -1 \\ 0 & 7 & 9 & -1 \\ 0 & 1 & 5 & 3 \\ 0 & 1 & 1 & -1 \end{bmatrix}$

II. $R_3 \longrightarrow 7R_3 - R_2$
$R_4 \longrightarrow 7R_4 - R_2$

$A \sim \begin{bmatrix} 2 & -1 & -3 & -1 \\ 0 & 7 & 9 & -1 \\ 0 & 0 & 26 & 22 \\ 0 & 0 & -2 & -6 \end{bmatrix}$

III. $R_4 \longrightarrow 13R_4 + R_3$

$A \sim \begin{bmatrix} 2 & -1 & -3 & -1 \\ 0 & 7 & 9 & -1 \\ 0 & 0 & 26 & 22 \\ 0 & 0 & 0 & -58 \end{bmatrix}$ $\therefore \rho(A) = 4$

c) $A = \begin{bmatrix} 2 & 1 & 0 & 0 \\ 1 & 2 & 1 & 0 \\ 0 & 1 & 2 & 1 \\ 1 & 0 & 1 & 2 \end{bmatrix}$

I. $R_2 \to 2R_2 - R_1$
$R_4 \to 2R_4 - R_1$

$$A \sim \begin{bmatrix} 2 & 1 & 0 & 0 \\ 0 & 3 & 2 & 0 \\ 0 & 1 & 2 & 1 \\ 0 & -1 & 2 & 1 \end{bmatrix}$$

II. $R_3 \to 3R_3 - R_2$
$R_4 \to 3R_4 + R_2$

$$A \sim \begin{bmatrix} 2 & 1 & 0 & 0 \\ 0 & 3 & 2 & 0 \\ 0 & 0 & 4 & 3 \\ 0 & 0 & 4 & 3 \end{bmatrix}$$

II. $R_4 \to R_4 - 2R_3$

$$A \sim \begin{bmatrix} 2 & 1 & 0 & 0 \\ 0 & 3 & 2 & 0 \\ 0 & 0 & 4 & 3 \\ 0 & 0 & 0 & 6 \end{bmatrix}$$

$\therefore \rho(A) = 4$

8/1

f) $A = \begin{bmatrix} 3 & 1 & 0 & 0 \\ 1 & 3 & 1 & 0 \\ 0 & 1 & 3 & 1 \\ 1 & 0 & 1 & 3 \end{bmatrix}$

$R_2 \to 3R_2 - R_1$
$R_4 \to 3R_4 - R_1$

$\sim \begin{bmatrix} 3 & 1 & 0 & 0 \\ 0 & 8 & 3 & 0 \\ 0 & 1 & 3 & 1 \\ 0 & -1 & 3 & 9 \end{bmatrix}$

$R_3 \to 8R_3 - R_2$; $R_4 \to 8R_4 + R_2$

$\sim \begin{bmatrix} 3 & 1 & 0 & 0 \\ 0 & 8 & 3 & 0 \\ 0 & 0 & 21 & 8 \\ 0 & 0 & 27 & 72 \end{bmatrix}$

$R_4 \to 21R_4 - 27R_3$

$\sim \begin{bmatrix} 3 & 1 & 0 & 0 \\ 0 & 8 & 3 & 0 \\ 0 & 0 & 21 & 8 \\ 0 & 0 & 0 & 936 \end{bmatrix}$

$\rho(A) = 4$

8|1 Consistent and Inconsistent system of LE

Consider a sys. of eqⁿ (or) $AX = B$, it is consistent if it has a solⁿ otherwise it is inconsistent.

Let $A = \begin{bmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \cdots & a_{2n} \\ a_{m1} & a_{m2} & a_{m3} & \cdots & a_{mn} \end{bmatrix}$, $X = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$, $B = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix}$

A matrix is formed by appending b the matrix A by an extra column consisting of the elements of B is called augment Matrix. It is denoted by $[A:B]$ and given by

$$[A:B] = \left[\begin{array}{cccc:c} a_{11} & a_{12} & \cdots & a_{1n} & b_1 \\ a_{21} & a_{22} & \cdots & a_{2n} & b_2 \\ a_{m1} & a_{m2} & \cdots & a_{mn} & b_m \end{array}\right]$$

If the sys. of eq^n are Consistent, then there are soln.

(i) Unique sol^n $\longrightarrow$ $\rho(A) = \rho(A:B) = \eta = n$
(ii) Infinite sol^n $\longrightarrow$ $\rho(A) = \rho(A:B) = \eta < n$

where n is no. of unknown variables in the given sys. of eq^{n}s.

And the sys. is inconsistent, if there is no sol^n i.e.

$\rho(A) \neq \rho(A:B)$

NOTE: The variables corresponding to the pivot colm are called basic variables and others are called free ".

Q. Sums

Test for consistency & solve.

1. $2x_1 - x_2 + 3x_3 = 1$; $-3x_1 + 4x_2 - 5x_3 = 0$; $x_1 + 3x_2 - 6x_3 = 0$

$$\Rightarrow [A:B] = \begin{bmatrix} 2 & -1 & 3 & : 1 \\ -3 & 4 & -5 & : 0 \\ 1 & 3 & -6 & : 0 \end{bmatrix}$$

$R_2 \rightarrow 2R_2 + 3R_1$; $R_3 \rightarrow 2R_3 - R_1$

$$\sim \begin{bmatrix} 2 & -1 & 3 & : 1 \\ 0 & 5 & -1 & : 3 \\ 0 & 7 & -15 & : -1 \end{bmatrix}$$

$R_3 \rightarrow 5R_3 - 7R_2$

$$\sim \begin{bmatrix} 2 & -1 & 3 & : 1 \\ 0 & 5 & -1 & : 3 \\ 0 & 0 & -68 & : -26 \end{bmatrix}$$

$\rho(A) = 3$; $\rho(A:B) = 3$; $n = 3$

$\therefore \rho(A) = \rho(A:B) = n = 3$

This system is consistent and has unique sol^n.

$AX = B$

$$2x_1 - x_2 + 3x_3 = 1 \quad \text{---} ①$$
$$0 + 5x_2 - x_3 = 3 \quad \text{---} ②$$
$$0 + 0 - 68x_3 = -26 \quad \text{---} ③$$

From ③, $x_3 = \frac{13}{34}$, From ②, $x_2 = \frac{115}{170} = \frac{23}{34}$

From ①, $x_1 = \frac{9}{34}$

2. $x + 2y + 2z = 1$
$2x + y + z = 2$
$3x + 2y + 2z = 3$
$y + z = 0$

$\Rightarrow [A:B] = \left[\begin{array}{ccc:c} 1 & 2 & 2 & 1 \\ 2 & 1 & 1 & 2 \\ 3 & 2 & 2 & 3 \\ 0 & 1 & 1 & 0 \end{array}\right]$

$R_2 \to 2R_2 - 2R_1$; $R_3 \to R_3 - 3R_1$

$\sim \left[\begin{array}{ccc:c} 1 & 2 & 2 & 1 \\ 0 & -3 & -3 & 0 \\ 0 & -4 & -4 & 0 \\ 0 & 1 & 1 & 0 \end{array}\right]$

$R_3 \to 3R_3 - 4R_2$; $R_4 \to 3R_4 + R_2$

$\sim \left[\begin{array}{ccc:c} 1 & 2 & 2 & 1 \\ 0 & -3 & -3 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$

$P(A) = 2$; $P(A:B) = 2$; $n = 3$
$P(A) = P(A:B) < n$
It is consistent and has ∞ sol^n.

* The pivot elements present in 1st & 2nd column.
x & y are basic and z is free variable

$x + 2y + 2z = 1$ —① ; $-3y - 3z = 0 \Rightarrow y + z = 0$ —②

Since z is a free variable, it has ∞ values; the values of x & y depend on z

From ②, $-z = +y$

From ①, $x = 1$

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ -z \\ z \end{bmatrix}.$$

9/1 Solution using Gauss elimination method

Procedure: ① write the Augmented matrix of given sys.

② Transform the Augmented matrix to regular form (The echelon ~~form~~ matrix will be a upper Δ matrix)

③ Write the matrix into linear eq^n's

④ By back substitution, obtain the unknowns.

SUMS (using Gauss elimination)

1. $x+y+z=9$; $2x-3y+4z=13$; $3x+4y+5z=40$

$$\Rightarrow \begin{bmatrix} 1 & 1 & 1 \\ 2 & -3 & 4 \\ 3 & 4 & 5 \end{bmatrix} = A \quad \begin{bmatrix} 9 \\ 13 \\ 40 \end{bmatrix} = B$$

$$[A:B] = \left[\begin{array}{ccc:c} 1 & 1 & 1 & 9 \\ 2 & -3 & 4 & 13 \\ 3 & 4 & 5 & 40 \end{array}\right]$$

$R_2 \to R_2 - 2R_1$; $R_3 \to R_3 - 3R_1$

$$\sim \begin{bmatrix} 1 & 1 & 1 & : & 9 \\ 0 & -5 & 2 & : & -5 \\ 0 & 1 & 2 & : & 13 \end{bmatrix}$$

$R_3 \to 5R_3 + R_2$

$$\sim \begin{bmatrix} 1 & 1 & 1 & : & 9 \\ 0 & -5 & 2 & : & -5 \\ 0 & 0 & 12 & : & 60 \end{bmatrix}$$

$P(A) = 3$; $P(A:B) = 3$
$n = 3$
$P(A) = P(A:B) = n = 3$

Unique sol^n.

$x + y + z = 9$ — ①
$-5y + 2z = -5$ — ②
$12z = 60$ — ③

From ③ , $z = 5$
From ② , $y = 3$
From ① , $x = 1$

$$\therefore \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 3 \\ 5 \end{bmatrix}$$

2. $5x + 3y + 7z = 4$; $3x + 26y + 2z = 9$; $7x + 2y + 10z$

$$\Rightarrow [A:B] = \begin{bmatrix} 5 & 3 & 7 & : & 4 \\ 3 & 26 & 2 & : & 9 \\ 7 & 2 & 10 & : & 5 \end{bmatrix}$$

$R_2 \to 5R_2 - 3R_1$; $R_3 \to 5R_3 - 7R_1$

$$\sim \begin{bmatrix} 5 & 3 & 7 & : & 4 \\ 0 & 121 & -11 & : & 33 \\ 0 & -11 & 1 & : & -3 \end{bmatrix}$$

$R_3 \longrightarrow 11R_3 + R_2$

$$\sim \begin{bmatrix} 5 & 3 & 7 & : & 4 \\ 0 & 121 & -11 & : & 33 \\ 0 & 0 & 0 & : & 0 \end{bmatrix}$$

$P(A) = P(A:B) = 2$

$n = 3$

Infinite sol^n

$5x + 3y + 7z = 4 \quad \text{—— } ①$

$121y - 11z = 33 \quad \text{— } ②$

From ②

$$11y - z = 3$$

$$y = \frac{3+z}{11}$$

From ①

$$5x + 3\left(\frac{3+z}{11}\right) + 7z = 4$$

$$5x = -\frac{9 - 3z}{11} - 7z + 4$$

$$5x = -\frac{9 - 3z - 77z + 44}{11}$$

$$x = \frac{-80z + 35}{55} = \frac{7 - 16z}{11}$$

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} (7-16z)/11 \\ (3+z)/11 \\ z \end{bmatrix}$$

3/5

3. $x+4y-z=5$; $x+y-6z=-12$; $3x-y-z=4$

$$\Rightarrow [A:B] = \begin{bmatrix} 1 & 4 & -1 & : & -5 \\ 1 & 1 & -6 & : & -12 \\ 3 & -1 & -1 & : & 4 \end{bmatrix}$$

$R_2 \to R_2 - R_1$; $R_3 \to R_3 - 3R_1$

$$\sim \begin{bmatrix} 1 & 4 & -1 & : & -5 \\ 0 & -3 & -5 & : & -7 \\ 0 & -13 & 2 & : & 19 \end{bmatrix}$$

$R_3 \to 3R_3 - 13R_2$

$$\sim \begin{bmatrix} 1 & 4 & -1 & : & -5 \\ 0 & -3 & -5 & : & -7 \\ 0 & 0 & 71 & : & 148 \end{bmatrix}$$

$\rho(A) = \rho(A:B) = n = 3$

Unique sol^n

$x+4y-z = -5$ — ①

$-3y-5z = -7$ — ②

$71z = 148$ — ③

From ③, $z = 148/71$

From ②, $y = -81/71$

From ①, $x = 117/71$

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 117/71 \\ -81/71 \\ 148/71 \end{bmatrix}$$

$y = \dfrac{-7+5z}{-3}$

$y = \dfrac{-7+5\left(\frac{148}{71}\right)}{-3}$

$= -\dfrac{497+740}{213}$

$= -\dfrac{243}{213}$

Q. Determi
eq^{n}s
$x+2y+$
has
$\Rightarrow [A:B]$
R_2
$\sim$
R_3
$\sim$
① (
②
③
5
$\Rightarrow$

Q Determine the value of a & b, so the system of eq^{n}s

$x+2y+3z=6$; $x+3y+5z=9$, $2x+5y+az=b$

has ① No (II) Unique (III) ∞

$$\Rightarrow [A:B] = \left[\begin{array}{ccc:c} 1 & 2 & 3 & 6 \\ 1 & 3 & 5 & 9 \\ 2 & 5 & a & b \end{array}\right]$$

$R_2 \to R_2 - R_1$; $R_3 \to R_3 - 2R_1$

$$\sim \left[\begin{array}{ccc:c} 1 & 2 & 3 & 6 \\ 0 & 1 & 2 & 3 \\ 0 & 1 & a-6 & b-12 \end{array}\right]$$

$R_3 \to R_3 - R_2$

$$\sim \left[\begin{array}{ccc:c} 1 & 2 & 3 & 6 \\ 0 & 1 & 2 & 3 \\ 0 & 0 & a-8 & b-15 \end{array}\right]$$

① $a=8$; $b \neq 15$
② $a \neq 8$; b any value
③ $a=8$; $b=15$

5 $2x+y+z=10$; $3x+2y+3z=18$; $x+4y+9z=16$

$$\Rightarrow [A:B] = \left[\begin{array}{ccc:c} 2 & 1 & 1 & 10 \\ 3 & 2 & 3 & 18 \\ 1 & 4 & 9 & 16 \end{array}\right]$$

13/1/26

$$[A:B] = \left[\begin{array}{ccc:c} 2 & 1 & 1 & 10 \\ 3 & 2 & 3 & 18 \\ 1 & 4 & 9 & 16 \end{array}\right]$$

$R_2 \to 2R_2 - 3R_1$; $R_3 \to 2R_3 - R_1$

$$\sim \left[\begin{array}{ccc:c} 2 & 1 & 1 & 10 \\ 0 & 1 & 3 & 6 \\ 0 & 7 & 17 & 22 \end{array}\right]$$

$R_3 \to R_3 - 7R_2$

$$\sim \left[\begin{array}{ccc:c} 2 & 1 & 1 & 10 \\ 0 & 1 & 3 & 6 \\ 0 & 0 & -4 & -20 \end{array}\right]$$

$P(A) = P(A:B) = n = 3$

Unique Sol^n

$2x + y + z = 10 \quad -①$

$y + 3z = 6 \quad -②$

$-4z = -20 \quad -③$

From ③

$z = 5$

From ②

$y = -9$

From ①

$x = 7$

$\therefore \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 7 \\ -9 \\ 5 \end{bmatrix}$

6- Determine a such that $x+y-z=1$; $2x+3y+az=3$; $x+ay+3z=+2$ ① No ② ∞ ③ Unique

$$\Rightarrow [A:B] = \left[\begin{array}{ccc:c} 1 & 1 & -1 & 1 \\ 2 & 3 & a & 3 \\ 1 & a & 3 & 2 \end{array}\right]$$

$R_2 \longrightarrow R_2 - 2R_1$; $R_3 \longrightarrow R_3 - R_1$

$$\sim \left[\begin{array}{ccc:c} 1 & 1 & -1 & 1 \\ 0 & 1 & a+2 & 1 \\ 0 & a-1 & 4 & 1 \end{array}\right]$$

$R_3 \longleftrightarrow R_2$

$$\sim \left[\begin{array}{ccc:c} 1 & 1 & -1 & 1 \\ 0 & a-1 & 4 & 1 \\ 0 & 1 & a+2 & 1 \end{array}\right]$$

$R_3 \longrightarrow R_3 + (a-1)R_2$

$$\sim \left[\begin{array}{ccc:c} 1 & 1 & -1 & 1 \\ 0 & a-1 & a+2 & 1 \\ 0 & 0 & 6-a^2-a & 2-a \end{array}\right]$$

$$\sim \left[\begin{array}{ccc:c} 1 & 1 & -1 & 1 \\ 0 & 1 & a+2 & 1 \\ 0 & 0 & (a+3)(a-2) & 2-a \end{array}\right]$$

① No : $a = -3$; ② ∞ : $a = 2$; ③ Unique : $a \neq -3, 2$

L U Decomposition

is an eq^n that express the Matrix A as product of 2 matrices.
Let A be a $m \times n$ matrix that can be reduced to echelon form without row interchange.
The matrix A can be written as $A = LU$ where L is lower Δ matrix with diagonal as 1 &

classmate
Date
Page

i.e. $A = LU$

$$= \begin{bmatrix} 1 & 0 & 0 \\ l_{21} & 1 & 0 \\ l_{31} & l_{32} & 1 \end{bmatrix} \begin{bmatrix} u_{11} & u_{12} & u_{13} \\ 0 & u_{22} & u_{23} \\ 0 & 0 & u_{33} \end{bmatrix}$$

Find L & U for A.

① $A = \begin{bmatrix} 3 & -7 & -2 & 2 \\ -3 & 5 & 1 & 0 \\ 6 & -1 & 0 & -5 \\ -9 & 5 & -5 & 12 \end{bmatrix}$ $L = \begin{bmatrix} 3 & & & \\ -3 & -2 & & \\ 6 & 10 & -1 & \\ 9 & -16 & -3 & -1 \end{bmatrix}$

$R_2 \to R_2 + R_1$; $R_3 \to R_3 - 2R_1$; $R_4 \to R_4 + 3R_1$

$$\sim \begin{bmatrix} 3 & -7 & -2 & 2 \\ 0 & -2 & -1 & 2 \\ 0 & 10 & 4 & -9 \\ 0 & -16 & -11 & 18 \end{bmatrix}$$

$R_3 \to R_3 + 5R_2$; $R_4 \to R_4 - 8R_2$

$$\sim \begin{bmatrix} 3 & -7 & -2 & 2 \\ 0 & -2 & -1 & 2 \\ 0 & 0 & -1 & 1 \\ 0 & 0 & -3 & 2 \end{bmatrix}$$

$R_4 \to R_4 - 3R_3$

$$\sim \begin{bmatrix} 3 & -7 & -2 & 2 \\ 0 & -2 & -1 & 2 \\ 0 & 0 & -1 & 1 \\ 0 & 0 & 0 & -1 \end{bmatrix} = U$$

I. Consider a sq. matrix A transforming to echelon form, the obtained echelon matrix is U.

II. In each step, before making the elements which are below the leading entry zero, divide the leading entry replace the same position in L by resultant elements.

Example: Find L & U.

1. $A = \begin{bmatrix} 1 & 1 & 1 \\ 4 & 3 & -1 \\ 1 & 1 & 3 \end{bmatrix}$ $L = \begin{bmatrix} 1 & & \\ 4 & -1 & \\ 1 & 0 & 2 \end{bmatrix}$

$R_2 \to R_2 - 4R_1$

$R_3 \to R_3 - R_1$

$\sim \begin{bmatrix} 1 & 1 & 1 \\ 0 & -1 & -5 \\ 0 & 0 & 2 \end{bmatrix} = U$ $L = \begin{bmatrix} 1 & 0 & 0 \\ 4 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$

$A = LU$

$= \begin{bmatrix} 1 & 0 & 0 \\ 4 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 \\ 0 & -1 & -5 \\ 0 & 0 & 2 \end{bmatrix} \Rightarrow \begin{bmatrix} 1 & 1 & 1 \\ 4 & 3 & -1 \\ 1 & 1 & 3 \end{bmatrix}$

2. $A = \begin{bmatrix} 1 & 1 & -1 \\ 2 & 3 & 5 \\ 3 & 2 & -3 \end{bmatrix}$ $L = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 3 & -1 & 1 \end{bmatrix}$

$R_2 \to R_2 - 2R_1$; $R_3 \to R_3 - 3R_1$

$\sim \begin{bmatrix} 1 & 1 & -1 \\ 0 & 1 & 7 \\ 0 & -1 & 0 \end{bmatrix}$ $R_3 \to R_3 + R_2$ $\sim \begin{bmatrix} 1 & 1 & -1 \\ 0 & 1 & - \\ 0 & 0 & \end{bmatrix} = U$

$$A = LU$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 3 & -1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 & -1 \\ 0 & 1 & 7 \\ 0 & 0 & 7 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 1 & -1 \\ 2 & 3 & 5 \\ 3 & 2 & -3 \end{bmatrix}$$

3. $A = \begin{bmatrix} 2 & 4 & -1 & 5 & -2 \\ -4 & -5 & 3 & -8 & 1 \\ 2 & -5 & -4 & 1 & 8 \\ -6 & 0 & 7 & -3 & 1 \end{bmatrix}$

$R_2 \longrightarrow R_2 + 2R_1$
$R_3 \longrightarrow R_3 - R_1$
$R_4 \longrightarrow R_4 + 3R_1$

$$L = \begin{bmatrix} 1 & 0 & 0 & 0 \\ -2 & 1 & 0 & 0 \\ 1 & -3 & 1 & 0 \\ -3 & 4 & 2 & 1 \end{bmatrix}$$

$$\sim \begin{bmatrix} 2 & 4 & -1 & 5 & -2 \\ 0 & 3 & 1 & 2 & -3 \\ 0 & -9 & -3 & -4 & 10 \\ 0 & 12 & 4 & 12 & -5 \end{bmatrix}$$

$R_3 \longrightarrow R_3 + 3R_2$
$R_4 \longrightarrow R_4 - 4R_2$

$$\sim \begin{bmatrix} 2 & 4 & -1 & 5 & -2 \\ 0 & 3 & 1 & 2 & -3 \\ 0 & 0 & 0 & 2 & 1 \\ 0 & 0 & 0 & 4 & 7 \end{bmatrix}$$

$R_4 \longrightarrow R_4 - 2R_3$

$$\sim \begin{bmatrix} 2 & 4 & -1 & 5 & -2 \\ 0 & 3 & 1 & 2 & -3 \\ 0 & 0 & 0 & 2 & 1 \\ 0 & 0 & 0 & 0 & 5 \end{bmatrix} = U$$

$A = LU$

$$= \begin{bmatrix} 1 & 0 & 0 & 0 \\ -2 & 1 & 0 & 0 \\ 1 & -3 & 1 & 0 \\ -3 & 4 & 2 & 1 \end{bmatrix} \begin{bmatrix} 2 & 4 & -1 & 5 & -2 \\ 0 & 3 & 1 & 2 & -3 \\ 0 & 0 & 0 & 2 & 1 \\ 0 & 0 & 0 & 0 & 5 \end{bmatrix}$$

4. $A = \begin{bmatrix} 2 & 1 & 0 \\ 4 & 3 & 1 \\ 6 & 5 & 2 \\ 8 & 7 & 3 \\ 10 & 9 & 4 \end{bmatrix}$ $L = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 2 & 1 & 0 & 0 & 0 \\ 3 & 2 & 1 & 0 & 0 \\ 4 & 3 & 0 & 1 & 0 \\ 5 & 4 & 0 & 0 & 1 \end{bmatrix}$

$R_2 \rightarrow R_2 - 2R_1$; $R_4 \rightarrow R_4 - 4R_1$
$R_3 \rightarrow R_3 - 3R_1$; $R_5 \rightarrow R_5 - 5R_1$

$$\sim \begin{bmatrix} 2 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 2 & 2 \\ 0 & 3 & 3 \\ 0 & 4 & 4 \end{bmatrix}$$

$R_3 \rightarrow R_3 - 2R_2$; $R_4 \rightarrow R_4 - 3R_2$
$R_5 \rightarrow R_5 - 4R_2$

$$\sim \begin{bmatrix} 2 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} = U$$

$A = LU$

$$= \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 2 & 1 & 0 & 0 & 0 \\ 3 & 2 & 1 & 0 & 0 \\ 4 & 3 & 0 & 1 & 0 \\ 5 & 4 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 2 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

16/1/26

Eigen values & vectors

Let A be a sq. matrix of order n & eigen vector of A is a non-0 vector such that $AX = \lambda X$ for some scalar λ where λ is called eigen value of matrix A. If there is a non-trivial sol^n $AX = \lambda X$, the vector X is called eigen vector corresponding to eigen value λ

Consider a unit sq. matrix of order n, then $|A - \lambda I| = 0$ gives the characteristic eq^n, the roots of this eq^n are eigen values. We can find eigen vector corresponding eigen value using $(A - \lambda I)X = 0$

NOTE: ① The sum of eigen values = Sum of the elements of main diagnol

② Tr(sq. matrix) is sum of its main elements
i.e. $Tr(A) = a_{11} + a_{22} + a_{33} + \ldots + a_{nn}$

③ The characteristic eq^n is a) For 2×2 : $\lambda^2 - Tr(A)\lambda + |A| = 0$

b) For 3×3 : $\lambda^3 - Tr(A)\lambda^2 + (M_{11} + M_{22} + M_{33})\lambda - |A| = 0$
where M_{ii} is the principal minor

Ex: Find eign value & vector of $A = \begin{bmatrix} 5 & 4 \\ 1 & 2 \end{bmatrix}$

Step 1: $A - \lambda I$

$$\begin{bmatrix} 5 & 4 \\ 1 & 2 \end{bmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix} \Rightarrow \begin{bmatrix} 5-\lambda & 4 \\ 1 & 2-\lambda \end{bmatrix}$$

Step 2: $|A - \lambda I| = 0$

$$(5-\lambda)(2-\lambda) - 4 = 0$$
$$10 - 2\lambda - 5\lambda + \lambda^2 - 4 = 0$$
$$-7\lambda + \lambda^2 + 6 = 0$$
$$\lambda^2 - 7\lambda + 6 = 0$$
$$(\lambda - 6)(\lambda - 1) = 0$$
$$\lambda = 6, 1$$

Step 3: Characteristic eqⁿ

$$\lambda^2 - \text{Tr}(A)\lambda + |A| = 0$$
$$\lambda^2 - 7\lambda + 6 = 0$$

Step 4: Find $x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$ when $\lambda = 6$ using $(A - \lambda I)x = 0$

$$A - \lambda I = \begin{bmatrix} -1 & 4 \\ 1 & -4 \end{bmatrix}$$

$$\begin{bmatrix} -1 & 4 \\ 1 & -4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 0$$

$$\begin{bmatrix} -x_1 + 4x_2 \\ x_1 - 4x_2 \end{bmatrix} = 0$$

$$-x_1 + 4x_2 = 0 \Rightarrow x_1 = 4x_2$$

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 4 \\ 1 \end{bmatrix}$$

$\lambda = 1$

$$A-\lambda I = \begin{bmatrix} 4 & 4 \\ 1 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 4 & 4 \\ 1 & 1 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 0$$

$$\begin{bmatrix} 4x_1 + 4x_2 \\ x_1 + x_2 \end{bmatrix} = 0$$

$$\Rightarrow x_1 = -x_2$$

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

Sums

① $\begin{bmatrix} 4 & -5 \\ 1 & -2 \end{bmatrix}$

$$|A-\lambda I| \Rightarrow \begin{bmatrix} 4 & -5 \\ 1 & -2 \end{bmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix} = \begin{bmatrix} 4-\lambda & -5 \\ 1 & -2-\lambda \end{bmatrix}$$

$$\Rightarrow \lambda^2 - 2\lambda - 3 = 0$$

$$(\lambda - 3)(\lambda + 1) = 0$$

$$\lambda = 3, -1$$

$(A-\lambda I)X = 0$ when $\lambda = 3$

$$\begin{bmatrix} 4-3 & -5 \\ 1 & -2-3 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 0$$

$$\begin{bmatrix} x_1 - 5x_2 \\ x_1 - 5x_2 \end{bmatrix} = 0 \qquad \Rightarrow x_1 = 5x_2$$

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} =$$

when

$$\begin{bmatrix} 4+ \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} 5x \\ x \end{bmatrix}$$

$3x$

$3x$

② $\begin{bmatrix} -5 \\ 2 \end{bmatrix}$

$A -$

λ

λ

$\begin{bmatrix}$

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 5 \\ 1 \end{bmatrix}$$

when $\lambda = -1$

$$\begin{bmatrix} A+1 & -5 \\ 1 & -2+1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 0$$

$$\begin{bmatrix} 3x_1 - 5x_2 \\ x_1 - 3x_2 \end{bmatrix} = 0$$

$x_1 = x_2$

$3x_1 - 5x_2 = 0$

$3x - 9x_2 = 0$

$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$

(2) $\begin{bmatrix} -5 & 2 \\ 2 & -2 \end{bmatrix}$

$$A - \lambda I = \begin{bmatrix} -5 & 2 \\ 2 & -2 \end{bmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix} = \begin{bmatrix} -5-\lambda & 2 \\ 2 & -2-\lambda \end{bmatrix}$$

$\lambda^2 + 7\lambda + 6 = 0$

$(\lambda+6)(\lambda+1) = 0$

$\lambda = -6, -1$

$\lambda = -6$

$$\begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 0$$

$x_1 + 2x_2 = 0$

$x_1 = -2x_2$

$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} -2 \\ 1 \end{bmatrix}$

$\lambda = -1$

$$\begin{bmatrix} -4 & 2 \\ 2 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 0$$

$-4x_1 + 2x_2 = 0$

$2x_1 = x_2$

$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 1 \\ \end{bmatrix}$

③ $\begin{bmatrix} 1 & 0 & -1 \\ 1 & 2 & 1 \\ 2 & 2 & 3 \end{bmatrix}$

$A - \lambda I \Rightarrow \begin{bmatrix} 1-\lambda & 0 & -1 \\ 1 & 2-\lambda & 1 \\ 2 & 2 & 3-\lambda \end{bmatrix}$

$\lambda^3 - tr(A)\lambda^2 + (M_{11} + M_{22} + M_{33})\lambda - |A| = 0$

$\Rightarrow \lambda^3 - 6\lambda^2 + 11\lambda - 6 = 0$

$= (\lambda-3)(\lambda-2)(\lambda-1) = 0$

$\lambda = 1, 2, 3$

$\lambda = 1$

$\begin{bmatrix} 0 & 0 & -1 \\ 1 & 1 & 1 \\ 2 & 2 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 0$

$-x_3 = 0 \Rightarrow x_3 = 0$

$\begin{bmatrix} -x_3 \\ x_1 + x_2 + x_3 \\ 2x_1 + 2x_2 + 2x_3 \end{bmatrix} = 0$

$\Rightarrow x_3 = 0$

$x_1 = -x_2$

$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}$

$\lambda = 2$

$\begin{bmatrix} -1 & 0 & -1 \\ 1 & 0 & 1 \\ 2 & 2 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 0$

$\begin{bmatrix} -x_1 - x_3 \\ x_1 + x_3 \\ 2x_1 + 2x_2 + x_3 \end{bmatrix} = 0$

$x_1 = x_3$

$3x_1 + 2x_2 = 0$

$x_1 = -\frac{2x_2}{3}$

$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -2 \\ +3 \\ +2 \end{bmatrix}$

$\lambda = 3$

$$\begin{bmatrix} -2 & 0 & -1 \\ 1 & -1 & 1 \\ 2 & 2 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 0$$

$$\begin{bmatrix} -2x_1 - x_3 \\ x_1 - x_2 + x_3 \\ 2x_1 + 2x_2 \end{bmatrix} = 0$$

$$x_1 = -x_2$$
$$x_1 = -\frac{x_3}{2}$$

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \rightarrow \begin{bmatrix} -1 \\ \pm 1 \\ +2 \end{bmatrix}$$

20/1/26 **Cramer's Rule**

Consider any 2 eqns

$$a_1x + b_1y + c_1z = 0$$
$$a_2x + b_2y + c_2z = 0$$
$$a_3x + b_3y + c_3z = 0$$

$$\Rightarrow \frac{x}{\begin{vmatrix} b_1 & c_1 \\ b_2 & c_2 \end{vmatrix}} = \frac{-y}{\begin{vmatrix} a_1 & c_1 \\ a_2 & c_2 \end{vmatrix}} = \frac{z}{\begin{vmatrix} a_1 & b_1 \\ a_2 & b_2 \end{vmatrix}} = k$$

④ $\begin{bmatrix} 1 & 0 & 0 \\ 1 & 2 & 0 \\ 2 & 2 & 3 \end{bmatrix}$

$A-\lambda I \rightarrow \begin{bmatrix} 1-\lambda & 0 & 0 \\ 1 & 2-\lambda & 0 \\ 2 & 2 & 3-\lambda \end{bmatrix}$

$\lambda^3 - tr(A)\lambda^2 + (M_{11}+M_{22}+M_{33})\lambda - |A| = 0$

$\lambda^3 - 6\lambda^2 + 11\lambda - 6 = 0$

$\lambda = 1, 3, 2$

$\lambda = 1$

$\begin{bmatrix} 0 & 0 & 0 \\ 1 & 1 & 0 \\ 2 & 2 & 2 \end{bmatrix} \Rightarrow$ Ignored

$x + y + 0z = 0$

$2x + 2y + 2z = 0$

$$\frac{x}{\begin{vmatrix} b_1 & c_1 \\ b_3 & c_3 \end{vmatrix}} = \frac{-y}{\begin{vmatrix} a_1 & c_1 \\ a_3 & c_3 \end{vmatrix}} = \frac{z}{\begin{vmatrix} a_1 & b_1 \\ a_3 & b_3 \end{vmatrix}} = k$$

$$\frac{x}{2} = \frac{-y}{0} = \frac{z}{0} = k$$

$(x, y, z) = (2k, 0k, 0k) \Rightarrow x = 2, y = 0, z = 0$

$\lambda = 2$

$\begin{bmatrix} -1 & 0 & 0 \\ +1 & 0 & 0 \\ 2 & 0 & 0 \end{bmatrix} \Rightarrow$ $-x + 0y + 0z = 0$

$+x + 0y + 0z = 0$

$2x + 2y + z = 0$

$$\frac{x}{\begin{vmatrix} 0 & 0 \\ 2 & 1 \end{vmatrix}} = \frac{-y}{\begin{vmatrix} +1 & 0 \\ 2 & 1 \end{vmatrix}} = \frac{z}{\begin{vmatrix} +1 & 0 \\ 2 & 2 \end{vmatrix}} = k$$

$(x, y, z) = (0k, \mp 1k, +2k) \Rightarrow x=0, \underline{y=\mp 1, z=+2}$

$\lambda = 3$

$$\begin{bmatrix} -2 & 0 & 0 \\ 1 & -1 & 0 \\ 2 & 2 & 0 \end{bmatrix} \Rightarrow \begin{matrix} -2x = 0 \\ x - y = 0 \\ 2x + 2y = 0 \end{matrix}$$

$$\frac{x}{\begin{vmatrix} -1 & 0 \\ 2 & 0 \end{vmatrix}} = \frac{-y}{\begin{vmatrix} 1 & 0 \\ 2 & 0 \end{vmatrix}} = \frac{z}{\begin{vmatrix} 1 & -1 \\ 2 & 2 \end{vmatrix}} = k$$

$(x, y, z) = (0k, 0k, 4k) \Rightarrow x=0, \underline{y=0, z=4}$

(5) $$\begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix}$$

$$A - \lambda I = \begin{bmatrix} 2-\lambda & -1 & 1 \\ -1 & 2-\lambda & -1 \\ 1 & -1 & 2-\lambda \end{bmatrix}$$

$\lambda^3 - tr(A)\lambda^2 + (M_{11} + M_{22} + M_{33})\lambda - |A| = 0$

$\lambda^3 - 6\lambda^2 + 9\lambda - 4 = 0$

$\lambda = \underline{4, 1, 1}$

$\lambda = 1$

$$\begin{bmatrix} 1 & -1 & 1 \\ -1 & 1 & -1 \\ 1 & -1 & 1 \end{bmatrix} \Rightarrow \begin{matrix} x-y+z=0 \quad -① \\ -x+y-z=0 \\ x-y+z=0 \end{matrix}$$

Since there are 2 eq, there will be 2 free variables, so cannot apply Cramer's Rule

Let x, y be the free vals. ① becomes $z = y - x$

$$\therefore \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} x \\ y \\ y-x \end{bmatrix} \Rightarrow x\begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} + y\begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$$

$\therefore$ Eigen values $= \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$

$\lambda = 4$

$$\begin{bmatrix} -2 & -1 & 1 \\ -1 & -2 & -1 \\ 1 & -1 & -2 \end{bmatrix} \Rightarrow \begin{matrix} -2x-y+z=0 \\ -x-2y-z=0 \\ x-y-2z=0 \end{matrix}$$

$$\frac{x}{\begin{vmatrix} -2 & -1 \\ -1 & -2 \end{vmatrix}} = \frac{-y}{\begin{vmatrix} -1 & -1 \\ 1 & -2 \end{vmatrix}} = \frac{z}{\begin{vmatrix} -1 & -2 \\ 1 & -1 \end{vmatrix}} = k$$

$$\frac{x}{3} = \frac{-y}{3} = \frac{z}{3} = k \Rightarrow (x, y, z) = (3k, -3k, 3k)$$

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$$

classmate
Date
Page

(6) $\begin{bmatrix} 3 & -2 & 4 \\ -2 & 6 & 2 \\ 4 & 2 & 3 \end{bmatrix}$

$|A-\lambda I| = \begin{bmatrix} 3-\lambda & -2 & 4 \\ -2 & 6-\lambda & 2 \\ 4 & 2 & 3-\lambda \end{bmatrix}$

$\lambda^3 - tr(A)\lambda^2 + (M_{11}+M_{22}+M_{33})\lambda - |A| = 0$

$\lambda^3 - 12\lambda^2 + 21\lambda + 98 = 0$

$\lambda = -2, 7, 7$

$\lambda = -2$

$\begin{bmatrix} 5 & -2 & 4 \\ -2 & 8 & 2 \\ 4 & 2 & 5 \end{bmatrix}$

$\lambda = 7$

$\begin{bmatrix} -4 & -2 & 4 \\ -2 & -1 & 2 \\ 4 & 2 & -4 \end{bmatrix}$

$\frac{x}{36} = \frac{-y}{-18} = \frac{z}{-36} = k$

$\frac{x}{0} = \frac{y}{0} = \frac{z}{0} = k$

$x, y, z = k(2, 1, -2)$

$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \\ -2 \end{bmatrix}$

$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} x \\ y \\ (2x+y)/2 \end{bmatrix}$

$\Rightarrow x\begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} + y\begin{bmatrix} 0 \\ 1 \\ 1/2 \end{bmatrix}$

$\therefore$ Eigen values $= \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ 1/2 \end{bmatrix}$

(7) $\begin{bmatrix} 3 & 1 & 4 \\ 0 & 2 & 6 \\ 0 & 0 & 5 \end{bmatrix}$

$\begin{bmatrix} 3-\lambda & 1 & 4 \\ 0 & 2-\lambda & 6 \\ 0 & 0 & 5-\lambda \end{bmatrix} = A - \lambda I$

$\lambda^3 - tr(A)\lambda^2 + (M_{11} + M_{22} + M_{33})\lambda - |A| = 0$

$\lambda^3 - 10\lambda^2 + (31)\lambda - 30 = 0$

$\lambda = 5, 3, 2$

$\lambda = 5$

$\begin{bmatrix} -2 & 1 & 4 \\ 0 & -3 & 6 \\ 0 & 0 & 0 \end{bmatrix}$

$\frac{x}{18} = \frac{-y}{-12} = \frac{z}{6} = k$

$(x, y, z) = (3, 2, 1)$

$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 3 \\ 2 \\ 1 \end{bmatrix}$

$\lambda = 3$

$\begin{bmatrix} 0 & 1 & 4 \\ 0 & -1 & 6 \\ 0 & 0 & 2 \end{bmatrix}$

$\frac{x}{10} = \frac{-y}{0} = \frac{z}{0} = k$

$(x, y, z) = 10k(-1, 0, 0)$

$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$

$\lambda = 2$

$\begin{bmatrix} 1 & 1 & 4 \\ 0 & 0 & 6 \\ 0 & 0 & 3 \end{bmatrix} \Rightarrow \frac{x}{6} = \frac{-y}{6} = \frac{z}{0} = k$

$(x, y, z) = 6k(1, -1, 0)$

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}$$

22/1/26 **Characteristic polynomial and minimal polynomial**

A characteristic polynomial (CP) of a sq. matrix A is defined as

$\chi(\lambda) = |A - \lambda I|$, where I is identity matrix of order same as A.

A minimal polynomial (MP), is a monic polynomial of lowest degree such that $m(A) = 0$ and it divides the characteristic polynomial.

Properties of M.P

- Every sq. matrix has unique MP.
- M.P of n divides characteristic polynomial.
- The fiers of factors $m(\lambda)$ are $\leq$ to those in CP
- If the eigen values are distinct then $m(\lambda)$ = CP.

Procedure

- Find the characteristic polynomial of a given matrix
- List out all the possible MP.
- Substitute A in each candidate polynomial starting from lowest degree
- The 1st poly MP for which $m(A) = 0$ is the MP for the given matrix.

Sums

Find CP & MP

1. $\begin{bmatrix} 5 & 1 \\ 3 & 7 \end{bmatrix}$

$$(A-\lambda I) = \begin{bmatrix} 5-\lambda & 1 \\ 3 & 7-\lambda \end{bmatrix}$$

$$\lambda^2 - tr(A)\lambda + |A| = 0$$
$$\lambda^2 - 12\lambda + 32 = 0$$
$$(\lambda-8)(\lambda-4)$$

$$m(\lambda) = (\lambda-8)(\lambda-4)$$
$$m(A) = 0$$
$$m(A) = (A-\lambda_1 I)(A-\lambda_2 I)$$

$$(A-8I)(A-4I) = 0$$

$$\begin{bmatrix} -3 & 1 \\ 3 & -1 \end{bmatrix}\begin{bmatrix} 1 & 1 \\ 3 & 3 \end{bmatrix} \Rightarrow \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} = \underline{O}$$

$\therefore$ M.P. $= (\lambda-8)(\lambda-4)$

2. $\begin{bmatrix} 4 & -1 \\ 1 & 2 \end{bmatrix}$

$$(A-\lambda I) = \begin{bmatrix} 4-\lambda & -1 \\ 1 & 2-\lambda \end{bmatrix}$$

$$\lambda^2 - 6\lambda + 9 = 0$$
$$(\lambda-3)(\lambda-3)$$

$m(A) = 0$

$(A-3I) \Rightarrow \begin{bmatrix} 1 & -1 \\ 1 & -1 \end{bmatrix} \neq 0$

$(A-3I)^2 \Rightarrow \begin{bmatrix} 1 & -1 \\ 1 & -1 \end{bmatrix}\begin{bmatrix} 1 & -1 \\ 1 & -1 \end{bmatrix}$

$= \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} = 0$

$\therefore M.P. = (\lambda-3)^2$

3. $\begin{bmatrix} 2 & 2 & -5 \\ 3 & 7 & -15 \\ 1 & 2 & -4 \end{bmatrix}$

$(A-\lambda I) = \begin{bmatrix} 2-\lambda & 2 & -5 \\ 3 & 7-\lambda & -15 \\ 1 & 2 & -4-\lambda \end{bmatrix}$

$\lambda^3 - 5\lambda^2 + 7\lambda - 3 = 0$

$(\lambda-3)(\lambda-1)^2$

$m(\lambda) = (\lambda-3)(\lambda-1)$ $\quad m(\lambda) = (\lambda-3)(\lambda-1)^2$

$m(A) = (A-3I)(A-I)$ $\quad m(A) = (A-3I)(A-I)(A-I)$

$\begin{bmatrix} -1 & 2 & -5 \\ 3 & 4 & -15 \\ 1 & 2 & -7 \end{bmatrix}\begin{bmatrix} 1 & 2 & -5 \\ 3 & 6 & -15 \\ 1 & 2 & -5 \end{bmatrix}$ $\quad \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}\begin{bmatrix} 1 & 2 & -5 \\ 3 & 6 & -15 \\ 1 & 2 & -5 \end{bmatrix}$

$= \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} = 0$ $\quad = 0$

$\therefore$ M.P. $= (\lambda-3)(\lambda-1)$

4. $\begin{bmatrix} 3 & 2 & -1 \\ 3 & 8 & -3 \\ 3 & 6 & -1 \end{bmatrix}$

$[A-\lambda I] = \begin{bmatrix} 3-\lambda & 2 & -1 \\ 3 & 8-\lambda & -3 \\ 3 & 6 & -1-\lambda \end{bmatrix}$

$\lambda^3 - 10\lambda^2 + 28\lambda - 24 = 0$

$(\lambda-2)^2(\lambda-6)$

$m(\lambda) = (\lambda-2)(\lambda-6)$ $\quad$ $m(\lambda) = (\lambda-2)^2(\lambda-6)$

$m(A) = (A-2I)(A-6I)$ $\quad$ $m(A) = (A-2I)(A-2I)(A-6I)$

$\begin{bmatrix} 1 & 2 & -1 \\ 3 & 6 & -3 \\ 3 & 6 & -3 \end{bmatrix}\begin{bmatrix} -3 & 2 & -1 \\ 3 & 2 & -3 \\ 3 & 6 & -7 \end{bmatrix}$ $\quad$ $\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}\begin{bmatrix} 1 & 2 & -1 \\ 3 & 6 & -3 \\ 3 & 6 & -3 \end{bmatrix}$

$= \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} = 0$ $\quad$ $= 0$

M.P. $= (\lambda-2)(\lambda-6)$

26 Rayliegh's power method

It is an iterative method to compute approx: value of the dominant eigen value (with largest magnitude) & it's corresponding eigen

vector for a given matrix.

Procedure: ① Consider matrix A and choose 1 initial non-zero vector say $X_0 = [x_1\ x_2\ x_3]^T$

② Compute AX_0. Equate $AX_0 = \lambda_1 X_1$ where λ_1 is the largest abs value in AX_0.

③ Compute $AX_1 = \lambda_2 X_2$. Repeat it till we get desired accurate x value.

Sums

Q. Find dominant eigen value & vector. Carry out 5 iterations.

1. $\begin{bmatrix} 2 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 0 & 2 \end{bmatrix} = A$; $X_0 = [1\ 0\ 0]^T$

$$AX_0 = \begin{bmatrix} 2 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 0 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \rightarrow \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix} = 2 \begin{bmatrix} 1 \\ 0 \\ 0.5 \end{bmatrix} = \lambda_1 X_1$$

$$AX_1 = \begin{bmatrix} 2 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 0 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 0.5 \end{bmatrix} = \begin{bmatrix} 2.5 \\ 0 \\ 2 \end{bmatrix} = 2.5 \begin{bmatrix} 1 \\ 0 \\ 0.8 \end{bmatrix} = \lambda_2 X_2$$

$$AX_2 = \begin{bmatrix} 2 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 0 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 0.8 \end{bmatrix} = \begin{bmatrix} 2.8 \\ 0 \\ 2.6 \end{bmatrix} = 2.8 \begin{bmatrix} 1 \\ 0 \\ 0.9285 \end{bmatrix} = \lambda_3 X_3$$

$$AX_3 = \begin{bmatrix} 2 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 0 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 0.9285 \end{bmatrix} = \begin{bmatrix} 2.9285 \\ 0 \\ 2.857 \end{bmatrix} = 2.9285 \begin{bmatrix} 1 \\ 0 \\ 0.9 \end{bmatrix} = \lambda_4 X_4$$

$$AX_4 = \begin{bmatrix} 2 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 0 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 0.9755 \end{bmatrix} = \begin{bmatrix} 2.9755 \\ 0 \\ 2.951 \end{bmatrix} = 2.9755 \begin{bmatrix} 1 \\ 0 \\ 0.9918 \end{bmatrix}$$

2. $\begin{bmatrix} 4 & 1 & -1 \\ 2 & 3 & -1 \\ -2 & 1 & 5 \end{bmatrix} = A$; $X_0 \begin{bmatrix} 1 \\ 0.8 \\ 0.8 \end{bmatrix}$

$$AX_0 = \begin{bmatrix} 4 & 1 & -1 \\ 2 & 3 & -1 \\ -2 & 1 & 5 \end{bmatrix} \begin{bmatrix} 1 \\ 0.8 \\ 0.8 \end{bmatrix} = \begin{bmatrix} 4 \\ 3.6 \\ 3.8 \end{bmatrix} = 4 \begin{bmatrix} 1 \\ 0.9 \\ 0.7 \end{bmatrix}$$

$$AX_1 = \begin{bmatrix} 4 & 1 & -1 \\ 2 & 3 & -1 \\ -2 & 1 & 5 \end{bmatrix} \begin{bmatrix} 1 \\ 0.9 \\ 0.7 \end{bmatrix} = \begin{bmatrix} 1.8823 \\ 1.7647 \\ 6.7058 \end{bmatrix} = 6.7058 \begin{bmatrix} 0.280 \\ 0.2631 \\ 1 \end{bmatrix}$$

$$AX_2 = \begin{bmatrix} 4 & 1 & -1 \\ 2 & 3 & -1 \\ -2 & 1 & 5 \end{bmatrix} \begin{bmatrix} 0.2807 \\ 0.2631 \\ 1 \end{bmatrix} = \begin{bmatrix} 0.3859 \\ 0.3508 \\ 5.8246 \end{bmatrix} = 5.8246 \begin{bmatrix} 0.0662 \\ 0.0602 \\ 1 \end{bmatrix}$$

$$AX_3 = \begin{bmatrix} 4 & 1 & -1 \\ 2 & 3 & -1 \\ -2 & 1 & 5 \end{bmatrix} \begin{bmatrix} +0.0662 \\ +0.0602 \\ 1 \end{bmatrix} = \begin{bmatrix} -0.6747 \\ -0.6867 \\ 5.1928 \end{bmatrix} = 5.1928 \begin{bmatrix} -0.1299 \\ -0.1322 \\ 1 \end{bmatrix}$$

$$AX_4 = \begin{bmatrix} 4 & 1 & -1 \\ 2 & 3 & -1 \\ -2 & 1 & 5 \end{bmatrix} \begin{bmatrix} -0.1299 \\ -0.1322 \\ 1 \end{bmatrix} = \begin{bmatrix} -1.6519 \\ -1.6566 \\ 4.6078 \end{bmatrix} = 4.6078 \begin{bmatrix} -0.3585 \\ -0.3594 \\ 1 \end{bmatrix}$$

3. $\begin{bmatrix} 1 & 3 & -1 \\ 3 & 2 & 4 \\ -1 & 4 & 10 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$

4. $\begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$

$3 \Rightarrow \begin{bmatrix} 1 & 3 & -1 \\ 3 & 2 & 4 \\ -1 & 4 & 10 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 3 \\ -1 \end{bmatrix} = 3 \begin{bmatrix} 1/3 \\ 1 \\ -1/3 \end{bmatrix} = 3 \begin{bmatrix} 0.3333 \\ 1 \\ -0.3333 \end{bmatrix}$

AX_0

$AX_1 = \begin{bmatrix} 1 & 3 & -1 \\ 3 & 2 & 4 \\ -1 & 4 & 10 \end{bmatrix} \begin{bmatrix} 0.3333 \\ 1 \\ -0.3333 \end{bmatrix} = \begin{bmatrix} 3.6666 \\ 1.6667 \\ 0.3337 \end{bmatrix} = 3.6666 \begin{bmatrix} 0.9090 \\ 0.4545 \\ -0.091 \end{bmatrix}$

$AX_2 = \begin{bmatrix} 1 & 3 & -1 \\ 3 & 2 & 4 \\ -1 & 4 & 10 \end{bmatrix} \begin{bmatrix} 1 \\ 0.4545 \\ -0.091 \end{bmatrix} = \begin{bmatrix} 2.2726 \\ 4.2731 \\ 1.7283 \end{bmatrix} = 4.2731 \begin{bmatrix} 0.5318 \\ 1 \\ 0.4044 \end{bmatrix}$

$AX_3 = \begin{bmatrix} 1 & 3 & -1 \\ 3 & 2 & 4 \\ -1 & 4 & 10 \end{bmatrix} \begin{bmatrix} 0.5318 \\ 1 \\ 0.4044 \end{bmatrix} = \begin{bmatrix} 3.1274 \\ 5.2134 \\ 7.5129 \end{bmatrix} = 7.5129 \begin{bmatrix} 0.4162 \\ 0.6939 \\ 1 \end{bmatrix}$

$AX_4 = \begin{bmatrix} 1 & 3 & -1 \\ 3 & 2 & 4 \\ -1 & 4 & 10 \end{bmatrix} \begin{bmatrix} 0.4162 \\ 0.6939 \\ 1 \end{bmatrix} = \begin{bmatrix} 1.498 \\ 6.6367 \\ 12.359 \end{bmatrix} = 12.359 \begin{bmatrix} 0.1212 \\ 0.5369 \\ 1 \end{bmatrix}$

$AX_5 = \begin{bmatrix} 1 & 3 & -1 \\ 3 & 2 & 4 \\ -1 & 4 & 10 \end{bmatrix} \begin{bmatrix} 0.1212 \\ 0.5369 \\ 1 \end{bmatrix} = \begin{bmatrix} 0.7321 \\ 5.4378 \\ 12.027 \end{bmatrix} = 12.027 \begin{bmatrix} 0.0608 \\ 0.4521 \\ 1 \end{bmatrix}$

4. $\begin{bmatrix} 2 & -1 & 0 \\ -1 & -2 & -1 \\ 0 & -1 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$

$AX_0 = \begin{bmatrix} 2 & -1 & 0 \\ -1 & -2 & -1 \\ 0 & -1 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 2 \\ -1 \\ 0 \end{bmatrix} = 2 \begin{bmatrix} 61 \\ -0.5 \\ 0 \end{bmatrix}$

$AX_1 = \begin{bmatrix} 2 & -1 & 0 \\ -1 & -2 & -1 \\ 0 & -1 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ -0.5 \\ 0 \end{bmatrix} = \begin{bmatrix} 2.5 \\ 0 \\ 0.5 \end{bmatrix} = 2.5 \begin{bmatrix} 1 \\ 0 \\ 0.2 \end{bmatrix}$

$AX_2 = \begin{bmatrix} 2 & -1 & 0 \\ -1 & -2 & -1 \\ 0 & -1 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 0.2 \end{bmatrix} = \begin{bmatrix} 2 \\ -1.2 \\ 0.4 \end{bmatrix} = 2 \begin{bmatrix} 1 \\ 0.6 \\ 0.2 \end{bmatrix}$

$AX_3 = \begin{bmatrix} 2 & -1 & 0 \\ -1 & -2 & -1 \\ 0 & -1 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ -0.6 \\ 0.2 \end{bmatrix} = \begin{bmatrix} 2.6 \\ 0 \\ 1 \end{bmatrix} = 2.6 \begin{bmatrix} 1 \\ 0 \\ 0.384 \end{bmatrix}$

$AX_4 = \begin{bmatrix} 2 & -1 & 0 \\ -1 & -2 & -1 \\ 0 & -1 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 0.3846 \end{bmatrix} = \begin{bmatrix} 2 \\ -1.384 \\ 0.7692 \end{bmatrix} = 2 \begin{bmatrix} 1 \\ -0.6923 \\ 0.3846 \end{bmatrix}$

$AX_5 = \begin{bmatrix} 2 & -1 & 0 \\ -1 & -2 & -1 \\ 0 & -1 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ -0.6923 \\ 0.3846 \end{bmatrix} = \begin{bmatrix} 2.6923 \\ 0 \\ 1.4615 \end{bmatrix} = 2.6923 \begin{bmatrix} 1 \\ 0 \\ 0.5428 \end{bmatrix}$